

Department Overview

- **Department Name:** Department of Computer Science
- **Faculty:** Faculty of Information & Communication Technology (FICT)
- **Accreditation:** The program is officially accredited by the **National Computing Education Accreditation Council (NCEAC)**.
- **History:** Founded in 2002 as one of the university's pioneer undergraduate programs during its initial inauguration. The department subsequently expanded to offer an MS degree program in 2006 and a PhD program in 2014.
- **Core Computational Focus:** Imparting foundational and technical knowledge on computer communication, networking, security, software development, database management systems, data sciences, and core programming.
- **Career Horizons:** Graduates transition into roles such as software engineers, data scientists, system administrators, and IT managers within industrial and governmental laboratories. The degree also serves as a multidisciplinary launchpad for careers in digital medicine, law, education, life sciences, and the humanities.

Vision and Objectives

- **Vision:** To provide a comprehensive computational education environment that inspires curiosity, creativity, and innovation for the enhancement of society in a positive and meaningful way.
- **Program Educational Objectives (PEOs):** Graduates are expected to:
 1. Responsibly practice systematic, disciplined, and quantifiable approaches to solve complex problems across computing and allied disciplines.
 2. Utilize skills and knowledge to serve industry and academia, positively impacting the socio-economic development of Pakistan.
 3. Demonstrate sustained learning and adaptability to evolving technological fields through continuous professional development and self-study.

1. Bachelor of Science in Computer Science (BSCS)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) / ICS with Mathematics and Physics from a recognized board (or equivalent qualification) with at least **60% marks**. Alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of **60% marks**.
- **Entrance Testing:** Candidates must clear an entrance exam administered by NTS.

Degree Requirements

- **Total Credit Hours:** 133 – 134
- **Total Courses:** 37 – 43
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour listings follow the format (Theory + Lab).

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	CS-110 / CS-110L	Introduction to ICT & Lab	None	2+1
	CS-114 / CS-114L	Programming Fundamentals & Lab	None	3+1
	HUM-175	English Composition & Comprehension	None	3+0
	MATHP-105	Calculus & Analytical Geometry	None	3+0
	PHY-205 / PHY-205L	Applied Physics & Lab	None	2+1
	HUM-101	Islamic Studies	None	2+0
2nd	CS-212 / L	Object Oriented Programming & Lab	Programming Fundamentals	3+1
	Hum-268	Communication & Presentation	English Composition & Comprehension	3+0

		Skills	n	
	—	CS Supporting I - Multivariable Calculus	Calculus & Analytical Geometry	3+0
	EE-102 / L	Digital Logic & Design & Lab	None	3+1
	MATHA-234	Discrete Structures	Applied Physics	3+0
3rd	CS-214 / L	Data Structure & Algorithms & Lab	Object Oriented Programming	3+1
	CS-334	Software Engineering	None	3+0
	CE-213 / L	Computer Organization & Assembly Language	Digital Logic & Design	3+1
	—	University Elective I – Introduction to Psychology	None	3+0
	MATHP-111	Linear Algebra	None	3+0
4th	CS-341 / L	Operating Systems & Lab	Data Structure & Algorithms	3+1

	TE-307 / L	Computer Networks & Lab	None	3+1
	CS-321	Theory of Automata	None	3+0
	—	CS Supporting II – Numerical Computing	None	3+0
	—	University Elective II – Human Resource Management	None	3+0
5th	HUM-266	Technical & Business Writing	None	3+0
	CS-421	Compiler Construction	Theory of Automata	3+0
	CS-412 / L	Artificial Intelligence & Lab	Discrete Structures	3+1
	CS-332 / L	Database Systems & Lab	Data Structure and Algorithm	3+1
	STAT-101	Probability and Statistics	None	3+0

6th	CS-424	Design & Analysis of Algorithm	Data Structure and Algorithm	3+0
	—	CS Elective I – Introduction to Big Data	None	2+1
	—	CS Elective II – Machine Learning	None	2+1
	—	CS Supporting III – Theory of Programming Languages	None	2+1
	—	University Elective III – Marketing	None	3+0
	—	University Elective IV – Social Service	None	0+0 (Pass/Fail)
7th	—	Final Year Project-I (FYDP-I)	Software Engineering, Technical Writing	0+3
	HUM-309	Professional Practices	None	3+0
	—	Parallel & Distributed	Operating	3+0

		Computing	Systems	
	—	Information Security	None	3+0
	—	CS Elective III – Cloud Computing	None	3+0
8th	—	Final Year Project-II (FYDP-II)	Final Year Project-I	0+3
	HUM-102	Pakistan Studies	None	2+0
	—	CS Elective IV – Advanced DBMS	None	3+0
	—	CS Elective V – Cryptography	None	3+0
	—	University Elective V – Wireless Networks	None	3+0

2. Master of Science (MS) in Computer Science

Admission Requirements

- **Academic Background:** 16 years of education or an equivalent qualification (such as a 4-year BS or B.E. degree) in a relevant discipline from an HEC-recognized university.
- **Minimum Marks:** At least **60% marks** under the annual grading system, or a minimum

CGPA of **2.5 out of 4.0**.

- **Testing:** A cumulative score of at least **50%** on an entry test conducted by the university or its designated authorized testing agency.

Degree Requirements

- **Total Credit Hours:** 30
- **Course Distribution:** 8 to 10 graduate-level courses.
- **Minimum Academic Standing:** CGPA $\geq$ 2.5

3. Doctor of Philosophy (PhD) in Computer Science

Admission Requirements

- **Academic Background:** An MS, M.Phil., or equivalent 18-year master's degree (completed after 16 years of basic schooling). The candidate's degree must feature a minimum of **30 credit hours** (consisting of 24 credit hours of graduate-level coursework + 6 credit hours of research thesis) from an HEC-recognized university or DAI (Degree Awarding Institution) in a relevant field.
- **Academic Performance:** Minimum of a 1st Division pass under the annual system, or an equivalent minimum CGPA of **3.0 out of 4.0**.
- **Testing Requirements:** Prior to confirmation of program admission, candidates must successfully pass either the standard **GRE (International) Subject Test** with a minimum 60th percentile score, or an authorized **GAT Subject Test** with at least **60% marks**.

Degree Requirements

- **Total Credit Hours:** 18
- **Course Distribution:** 6 to 8 advanced doctoral courses, alongside mandatory dissertation work.
- **Minimum Academic Standing:** CGPA $\geq$ 2.5